Run report: E6, yield of a pushed-forward factor base

2026-10-12, 02:27–02:30 UTC. research/isogeny_conductor_gap_round5_20261012.

Runner examples/isogeny_gap_pushforward_yield.rs (commit cb33477a1), 20,000 targets per curve; cloud container, no wall times reported. **Records** e6/records.jsonl: 112 arm records on 15 curves (crater, floor, floor model of five 22-bit round-1 pairs with h = 12, 18, 50, 72, 200), bases of the 32 and 64 smallest abscissae outside E[h]; arms direct and quotient by C2, C4, E[2] (where rational) and $E(F_p)[h]$. Every decomposition lifted back to E and checked: 0 failures.

subgroup	|H|	cases	quotient ÷ direct	95 % interval
C2	2	30	1.81	1.64 to 1.99
C4	4	16	4.08	3.52 to 4.75
E[2]	4	6	3.57	2.76 to 4.60
E[h]	12	6	8.43	6.00 to 11.86
E[h]	18	6	21.87	17.70 to 27.01
E[h]	50	6	44.07	39.24 to 49.50
E[h]	72	6	79.57	67.54 to 93.73
E[h]	200	6	139.39	103.62 to 187.50

Derived: weighted least squares of ln(ratio) on ln|H|, exponent 1.001 ± 0.036. F4 (one random target per arm): equal cost at matched columns on the 106 unsolvable targets (136,476–145,630 multiplications at c = 32, 581,788–600,510 at c = 64); 483,561 and 2,991,006–6,313,667 on the 6 solvable ones.

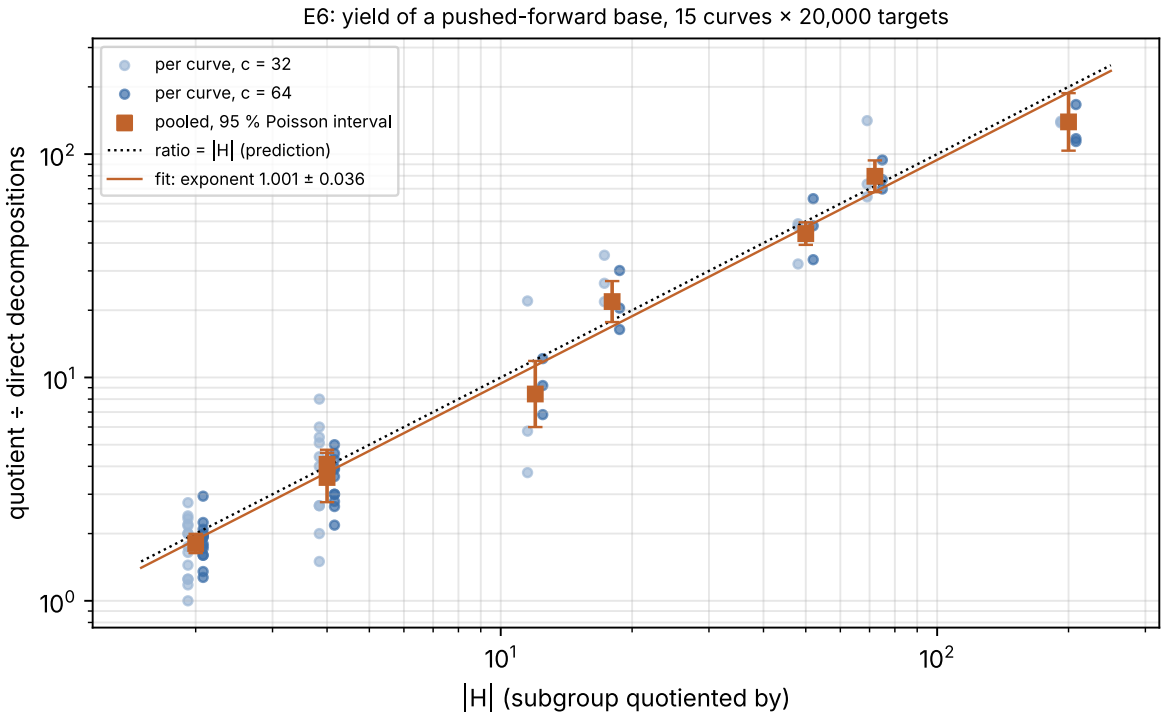

Figure 1: Decompositions on E/H ÷ on E over the same targets and columns, per curve and pooled (95 % Poisson intervals), against |H|. Records e6/records.jsonl.